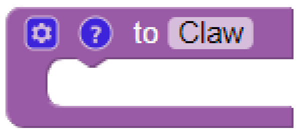
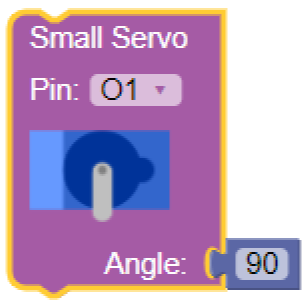
MicrokARts Coding Exercises

Coding Challenge 1

Grab a function block and rename it **"Claw"**

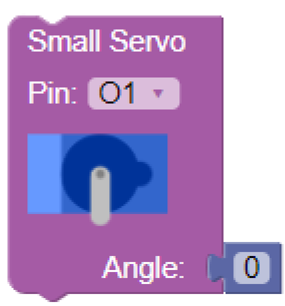


Grab a **"Small Servo"** block from the "Motors" folder and place it inside the "Claw" function block. Change the angle of the servo to 90.



Grab a <u>delay block</u> from the "Logic" folder and place it under the small servo block. Change the value to 2000 milliseconds (2 seconds).

Grab a second <u>Small Servo block</u> and place it under your "delay" block. Set the angle to 0.



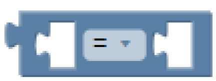
Your code will make your claw open, wait 2 seconds, and then close. Press play on the simulator and see if it behaves as you expect.

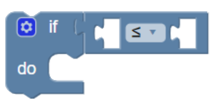
Coding Challenge 2

Grab a function block and rename it **"Nightlight."**

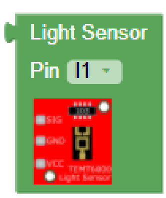
Grab an "If" block from the "Logic" folder and place it inside your Nightlight function block.

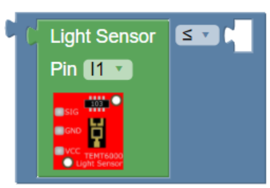
Grab a comparison block from the "Logic" folder and connect the ends with the puzzle pieces together. Change the "=" sign to a "≤" sign





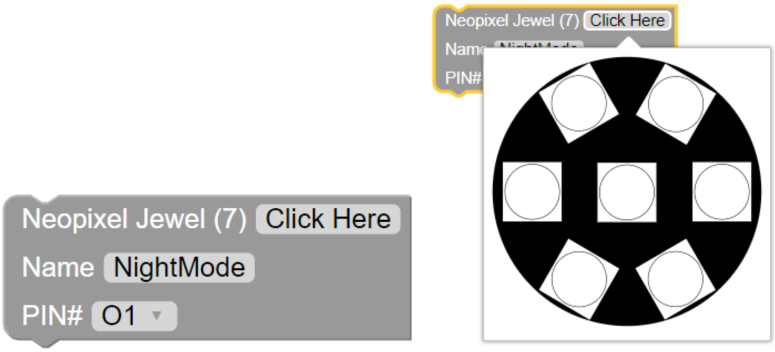
Grab a Light Sensor block from the "Sensors" folder and place it on the left side of the comparison block.





Go to the "Math" folder and grab a number block. Place it on the right side of the comparison block. This means that your code will execute whatever is in the "Do" part of the block IF the light sensor reads a value less than the number on the right. In order to get the correct number, **use Laboratory mode** on your IoT Board.

Grab a "Neopixel Jewel (7)" block from the "Lights" folder and place it in the "Do" part of the "If" block. Click your mouse on the part that says "click here" to bring up the image of the Neopixel. If you click on the gray parts, you will notice them change color. Turn each of them white. Finally, change the name of your Light to "NightMode"



Your code should be set to turn you lights white IF the light sensor reads a value less than the number you input. Press play on the simulator and see if it behaves as you expect.

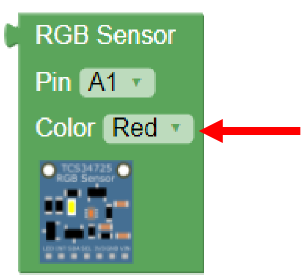
Coding Challenge 3

Grab a <u>function block</u> and rename it **"ColorPicker."**

Grab an <u>"If" block</u> and place it inside your ColorPicker block.

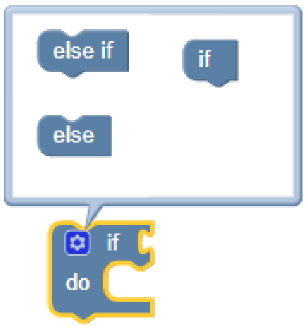
Grab a <u>comparison block</u> and connect the ends with the puzzle pieces together. Change the "=" sign to a "≥" sign

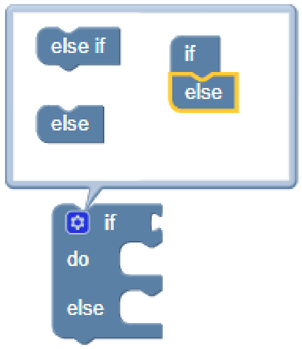
Grab an <u>RGB Sensor block</u> from the "Sensors" folder and place it on the left side of the comparison block. Make sure the color it is sensing is set to red.



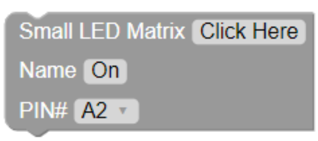
Go to the "Math" folder and grab a number block. Place it on the right side of the comparison block and change its value to something high (such as 200).

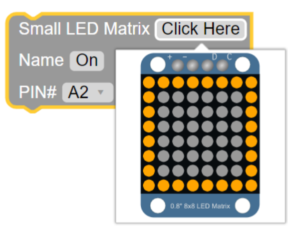
Click the gear located in the top left of the "If" block and connect the "Else" block to the bottom of the "If" block. You'll notice the shape of the "If" block will change. Anything that goes in this section of the block will execute IF the RGB sensor reads anything ELSE that is not inside the comparison block





Grab a "Small LED Matrix" block from the "Lights" folder and place it in the "Do" part of the "If" block. Click your mouse on the part that says "Click Here" to bring up the image of the LED matrix. Turn each of the outer-most LEDs on. Change the name from "default_bmp" to "On"





Grab another "Small LED Matrix" block and place it in the "Else" part of the "If" block. Click your mouse on the part that says "click here" to bring up the image of the LED Matrix. Make sure that each of them is gray. Change the name from "default_bmp" to "Off"

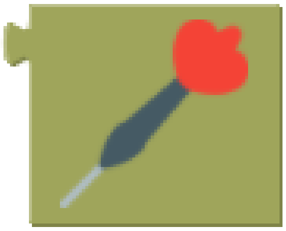
Your code should be set to turn the lights on IF the RGB sensor reads a high red value, ELSE, it will turn off. Press play on the simulator and see if it behaves as you expect.

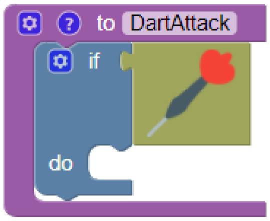
Coding Challenge 4

Grab a <u>function block</u> and rename it **"DartAttack"**

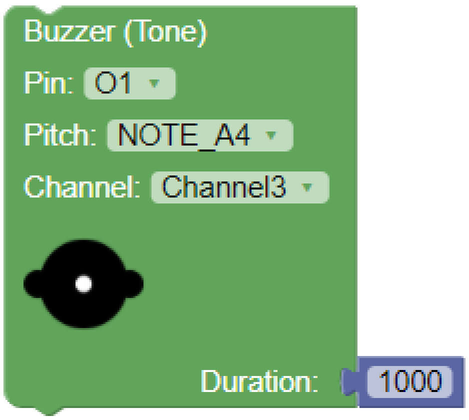
Grab an <u>"If" block</u> and place it inside your DartAttack block.

Grab an <u>AR Dart</u> block from the "AR" folder and connect it to the "If" block.





Grab a <u>Buzzer block</u> from the "Buzzers" folder and place it in the "do" part of the "If" block. Change the pitch to any of the available options. Change the duration to 1 second (1000 milliseconds).



You code should be set to make a buzzer noise IF your MicrokARts gets hit by a dart. Check the simulator to see if you hear the buzzer when you click the AR Dart button at the bottom!